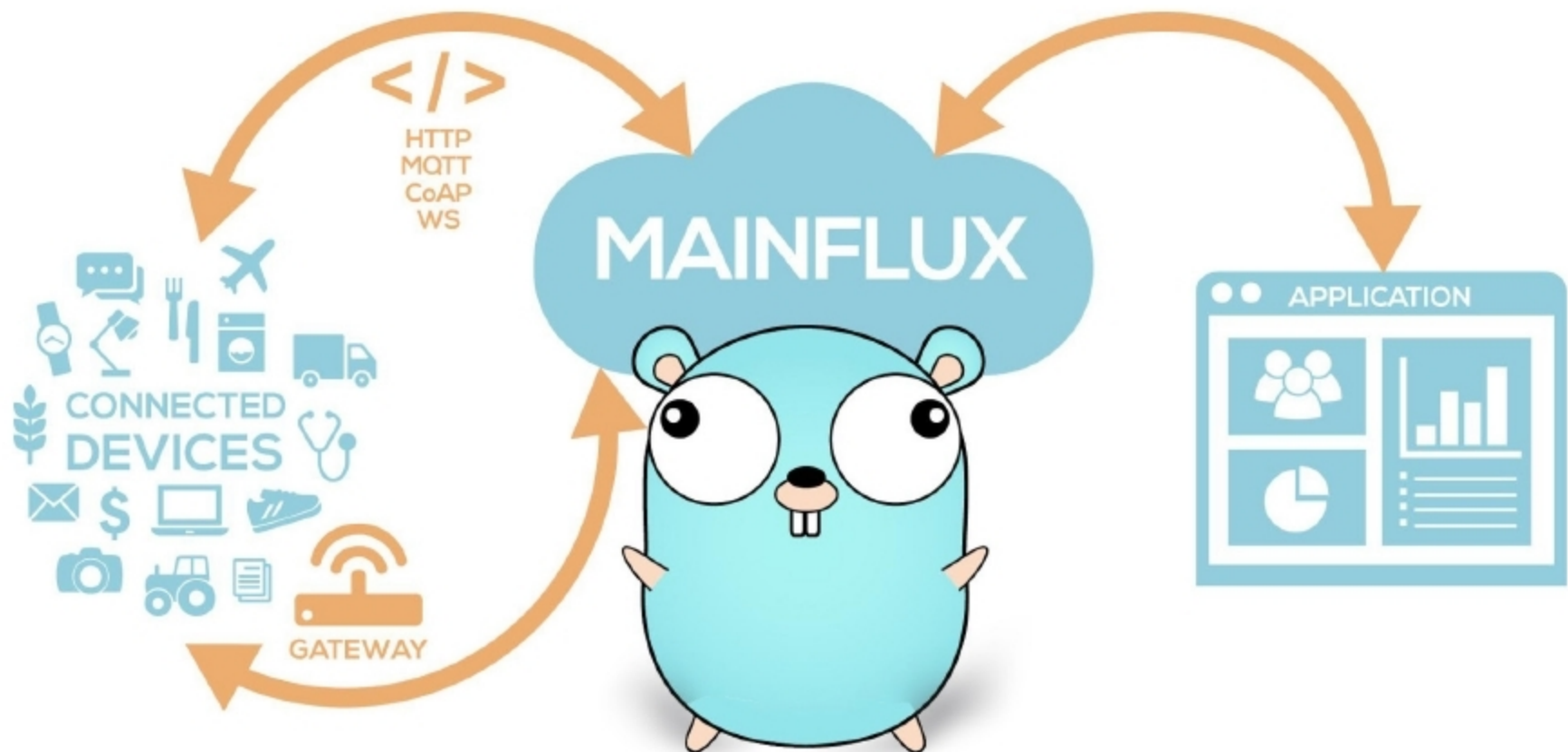


Exploring Mainflux and Its Usage

Mainflux is a highly secure, open source and patent-free IoT cloud platform developed in the GO programming language. It is based on a set of microservices. It accepts connections over multiple network protocols like HTTP, MQTT, Websocket, CoAP, etc, making a seamless bridge between them. Mainflux is used as IoT middleware, on top of which advanced IoT solutions can be constructed.



Mainflux is used as the IoT middleware for constructing complex IoT solutions. It makes use of south bound API to connect devices and northbound API to connect applications, and then it permits message routing between them. The whole bottom-to-top system (device, Mainflux platform, and application) bundled together is called a ‘vertical solution’.

From Figure 1, we can understand that Mainflux works as middleware that does a lot of heavy lifting. The product designers have only to focus

on connected devices (physical design, firmware, etc) and the end user application (GUI that controls the device), as the Mainflux block can be reused. Mainflux sends messages between connected devices and applications in a highly secure manner, also offering many other services needed for these devices and applications. It offers multi-protocol device-agnostic message relay with distributed time-series data storage, and multi-user, multi-tenant device and application management middleware.

Architecture

Mainflux works as middleware, which means it employs a set of servers that provide several functionalities and services used in developing IoT applications.

The features of Mainflux are:

- Messaging bridge (HTTP Sender, MQTT and WS, CoAP)—relays messages between devices and applications
- Systems manager (Manager)
- Device manager (Device)—accepts device connections on southbound interface

